

```

#Om Bhujbal 22CV007
# Program to Determine the bearing capacity of soil with water table

# Input values
BulkDensity = float(input("Enter the value of Bulk Density of soil (kN/m³): "))
SatDensity = float(input("Enter the value of Saturated Density of soil (kN/m³): "))
WaterDensity = float(input("Enter the unit Weight of Water (kN/m³): "))

Df = float(input("Enter the value of depth of footing Df (m): "))
Dw = float(input("Enter the value of water table above footing level Dw (m): "))

# Handle potential empty input for Dw1
while True:
    dw1_input = input("Enter the value of Water table below the level of footing ")
    try:
        Dw1 = float(dw1_input)
        break # Exit the loop if conversion is successful
    except ValueError:
        print("Invalid input. Please enter a valid number for Dw1.")

B = float(input("Enter the value of width of footing B (m): "))

Nq = float(input("Enter the value of Nq: "))
N = float(input("Enter the value of N (bearing capacity factor): "))

# Submerged density
SubDensity = SatDensity - WaterDensity
print("Submerged Weight of soil is:", SubDensity)

# ----- CASE A -----
print("\nCASE A: Water table at ground surface")
qu = (SubDensity * Df * Nq) + (0.5 * B * SubDensity * N)
print("The value of ultimate bearing capacity of soil is:", qu)

Rw = 0.5 + 0.5 * (Dw / B)
print("The value of Rw is:", Rw)

Rw1 = 0.5 + 0.5 * (Dw1 / B)
print("The value of Rw1 is:", Rw1)

qu = (BulkDensity * Df * Nq * Rw) + (0.5 * B * BulkDensity * N * Rw1)
print("The approximate value of ultimate bearing capacity is:", qu)

# ----- CASE B -----
print("\nCASE B: Water table at base of footing")
qu = (BulkDensity * Df * Nq) + (0.5 * B * SubDensity * N)
print("The value of ultimate bearing capacity is:", qu)

Rw = 0.5 + 0.5 * (Dw / B)
print("The value of Rw is:", Rw)

Rw1 = 0.5 + 0.5 * (Dw1 / B)
print("The value of Rw1 is:", Rw1)

qu = (BulkDensity * Df * Nq * Rw) + (0.5 * B * BulkDensity * N * Rw1)
print("The approximate value of ultimate bearing capacity is:", qu)

# ----- CASE C -----
print("\nCASE C: Water table below base of footing")

```

```

while True:
    x_input = input("Enter the value of depth of water below footing (x in m): ")
    try:
        x = float(x_input)
        break
    except ValueError:
        print("Invalid input. Please enter a valid number for x.")

qu = (BulkDensity * Df * Nq) + (0.5 * B * ((BulkDensity * x) + (SubDensity * (B -
print("The value of ultimate bearing capacity is:", qu)

Rw = 0.5 + 0.5 * (Dw / B)
print("The value of Rw is:", Rw)

Rw1 = 0.5 + 0.5 * (Dw1 / B)
print("The value of Rw1 is:", Rw1)

qu = (BulkDensity * Df * Nq * Rw) + (0.5 * B * BulkDensity * N * Rw1)
print("The approximate value of ultimate bearing capacity is:", qu)

```

```

Enter the value of Bulk Density of soil (kN/m³): 18
Enter the value of Saturated Density of soil (kN/m³): 20
Enter the unit Weight of Water (kN/m³): 10
Enter the value of depth of footing Df (m): 2
Enter the value of water table above footing level Dw (m): 0
Enter the value of Water table below the level of footing Dw1 (m): 0
Enter the value of width of footing B (m): 3
Enter the value of Nq: 33
Enter the value of N (bearing capacity factor): 34
Submerged Weight of soil is: 10.0

```

```

CASE A: Water table at ground surface
The value of ultimate bearing capacity of soil is: 1170.0
The value of Rw is: 0.5
The value of Rw1 is: 0.5
The approximate value of ultimate bearing capacity is: 1053.0

```

```

CASE B: Water table at base of footing
The value of ultimate bearing capacity is: 1698.0
The value of Rw is: 0.5
The value of Rw1 is: 0.5
The approximate value of ultimate bearing capacity is: 1053.0

```

```

CASE C: Water table below base of footing
Enter the value of depth of water below footing (x in m): 1
The value of ultimate bearing capacity is: 3126.0
The value of Rw is: 0.5
The value of Rw1 is: 0.5
The approximate value of ultimate bearing capacity is: 1053.0

```

```

#0m Bhujbal 22CV007
# Program 2: To find the ultimate load carrying capacity of pile
UCS = float(input("Enter the value of UCS of soil:"))
Cu = UCS / 2
B = float(input("Enter the value of dimension of pile:"))

```

```

L = float(input("Enter the length of pile:"))
Alpha = float(input("Enter the value of adhesion factor:"))
Nc = float(input("The value of Nc: "))

Ab = B * B
print("The Base area of footing is:", Ab)

As = 4 * B * L
print("The value of cohesion of soil is:", Cu)

Qpu = Cu * Nc * Ab
print("Qpu:", Qpu)

Qf = Alpha * Cu * As
print("Qf:", Qf)

Qu = Qpu + Qf
print("The value of load carrying capacity of pile is (Qu):", Qu)

```

```

Enter the value of UCS of soil:75
Enter the value of dimension of pile:0.45
Enter the length of pile:15
Enter the value of adhesion factor:0.8
The value of Nc: 9
The Base area of footing is: 0.2025
The value of cohesion of soil is: 37.5
Qpu: 68.34375
Qf: 810.0
The value of load carrying capacity of pile is (Qu): 878.34375

```

```

#Om Bhujbal 22CV007
# Program 3: To Determine the bearing capacity of soil with water table (multiple
BulkDensity = float(input("Enter the value of Bulk Density of soil:"))
SatDensity = float(input("Enter the value of Saturated Density of soil:"))
WaterDensity = float(input("Enter the unit Weight of Water:"))
Df = float(input("Enter the value of depth of footing:"))
B = float(input("Enter the value of width of footing:"))
Nq = float(input("Enter the value of Nq:"))
N_Gamma = float(input("Enter the value of N gamma (N):"))

SubDensity = SatDensity - WaterDensity
print("Submerged Weight of soil is:", SubDensity)

M = int(input("Number of data values of Water table above footing level: "))
N = int(input("Number of data values of Water table below footing level: "))

Dw = []
Dw1 = []

for i in range(1, M+1):
    Depth_Dw = float(input("Enter the value of water table above footing level me
    Dw.append(Depth_Dw)
    Rw = 0.5 + 0.5 * (Depth_Dw / B)
    print("The value of Rw is:", Rw)

for j in range(1, N+1):
    Depth_Dw1 = float(input("Enter the value of water table below footing level m
    Dw1.append(Depth_Dw1)
    Rw1 = 0.5 + 0.5 * (Depth_Dw1 / B)

```

```

print("The value of Rw1 is:", Rw1)

qu = (BulkDensity * Df * Nq * Rw) + (0.5 * 0.8 * B * BulkDensity * N_Gamma * Rw1)
print("qu: ", qu, "kN/m^2")

```

```

Enter the value of Bulk Density of soil:18
Enter the value of Saturated Density of soil:20
Enter the unit Weight of Water:10
Enter the value of depth of footing:2
Enter the value of width of footing:3
Enter the value of Nq:33
Enter the value of N gamma (N):34
Submerged Weight of soil is: 10.0
Number of data values of Water table above footing level: 3
Number of data values of Water table below footing level: 3
Enter the value of water table above footing level measured w.r.t. ground (m):0.5
The value of Rw is: 0.5
Enter the value of water table above footing level measured w.r.t. ground (m):0.6666666666666666
The value of Rw is: 0.6666666666666666
Enter the value of water table above footing level measured w.r.t. ground (m):0.8333333333333333
The value of Rw is: 0.8333333333333333
Enter the value of water table below footing level measured w.r.t. ground (m):0.5
The value of Rw1 is: 0.5
Enter the value of water table below footing level measured w.r.t. ground (m):0.5
The value of Rw1 is: 0.5
Enter the value of water table below footing level measured w.r.t. ground (m):0.6666666666666666
The value of Rw1 is: 0.6666666666666666
qu: 1479.6 kN/m^2

```